Inverse design of a suspended SiN nanobeam cavity for a ^{87}Rb–waveguide photon interface

Repository: atomTrap — automated inverse-design workflow (inverse_design/ package, Tidy3D autograd adjoint optimization). Run: invDesResults/full, 42 recorded iterations, 3.29 h optimization time, ≈67 FlexCredits total.

Summary

The hand-tuned baseline (design_1 from cavities_780.ipynb) was converted into a fully differentiable parameterization — 86 per-cell lattice constants, 172 per-hole scale factors, and 20 closed-spline shape control points (278 parameters) — and optimized against a frequency-domain surrogate of the physical photon-interface efficiency $\eta = (\kappa_{wg}/\kappa_{tot}) \cdot C/(C+1) \cdot \eta_{fiber}$, with resonance targeting at the ^{87}Rb D2 line (384.23 THz) and fabrication constraints (50 nm hole-hole gaps, 70 nm side rails, 40 nm curvature radius). Every reported physical number below comes from the repo's pre-existing time-domain verification pipeline (ResonanceFinder Q, per-face directional Q, mode volume), applied identically to baseline and optimized designs.

Headline result: the verified end-to-end collection efficiency at a realistic atom position (250 nm above the beam surface) improved 3.4×, from η = 0.083 to η = 0.278, while the resonance moved from 1.01 THz off the Rb D2 target to 0.29 THz — within thermal-tuning range. Cooperativity at the atom doubled (C = 21 → 45).

1. Verified performance vs. baseline

metric	baseline	optimized	change
f_res (THz) [target 384.23]	385.24	383.94	Δ: 1.01→0.29 THz
η (atom @250 nm)	0.083	0.278	3.36×
C (atom @250 nm)	21.2	45.0	2.12×
$\beta = \kappa_{wg}/\kappa_{tot}$	0.087	0.287	3.28×
Q (loaded)	1137	1778	1.56×
$\kappa_{tot}/2\pi$ (GHz)	339	216	—
$g/2\pi$ @250 nm (GHz)	3.30	3.84	1.16×
V_mode ($(\lambda/n)^3$, legacy conv.)	0.66	0.61	—
Purcell ($3Q/4\pi^2V$)	130	222	1.70×

Note on the baseline: the notebook-recorded Q = 1.33×10^6 for design_1 is not reproducible from the notebook's stored parameters (its 0.5 µm left-mirror lattice has a/λ ≈ 0.64 at 780 nm, far above the first TE gap — classic notebook cell-edit drift). The stored parameter set is the only reproducible 'current best design'; it was re-evaluated with the identical pipeline used for the optimized design.

2. The optimized design

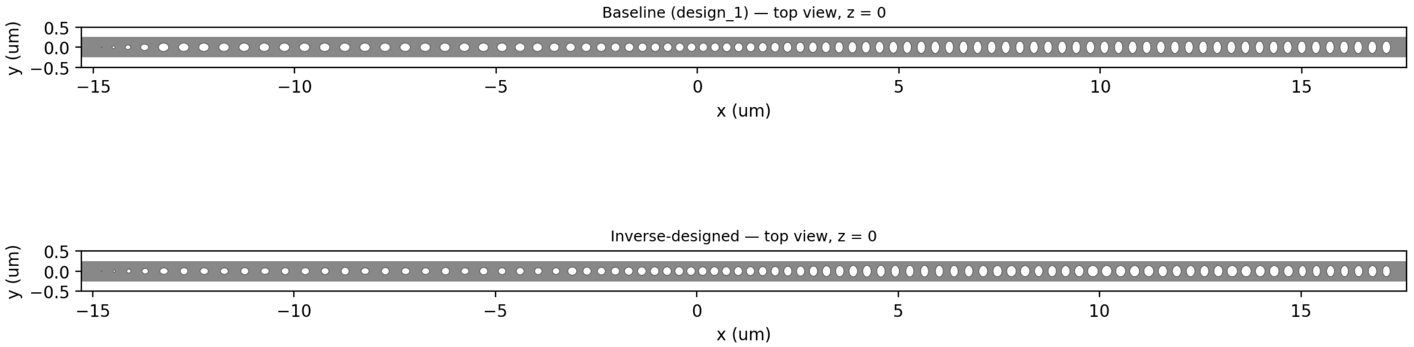

Fig. 1 — Top view (z = 0): baseline vs. optimized. Output waveguide at left (5-hole taper + 10 mirror cells); 40-cell defect; 30 back-mirror cells at right.

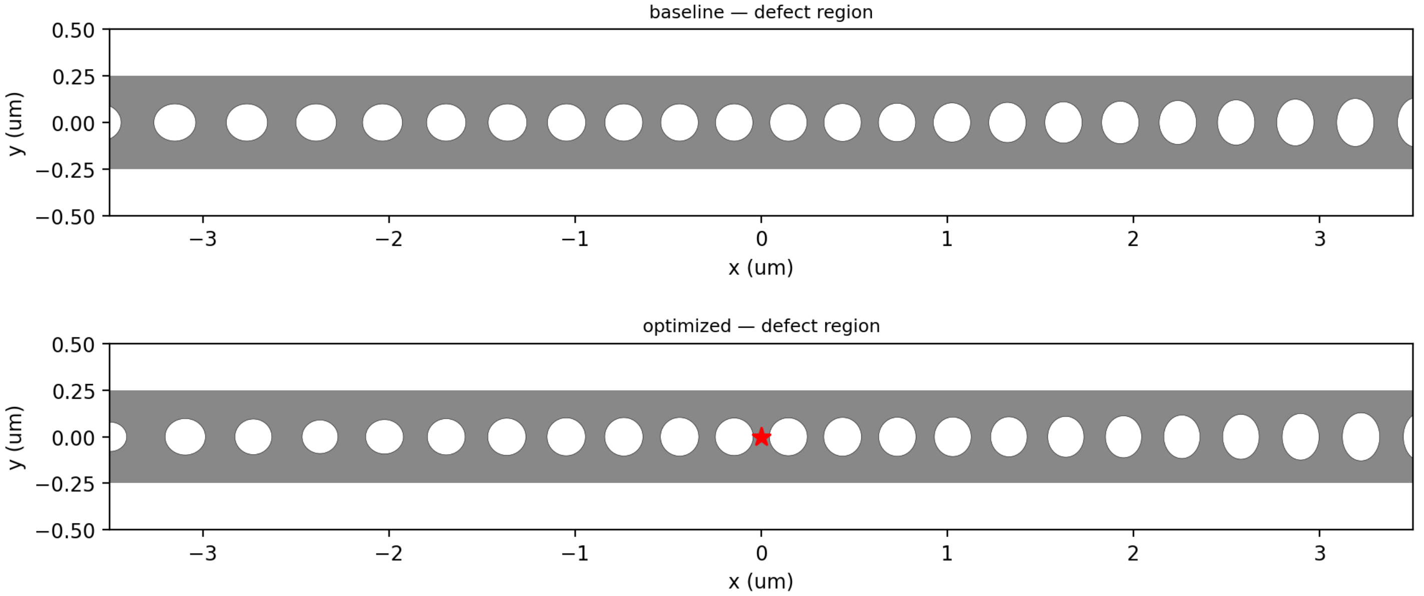

Fig. 2 — Defect region (red star: dipole/atom x-position at the central dielectric antinode).

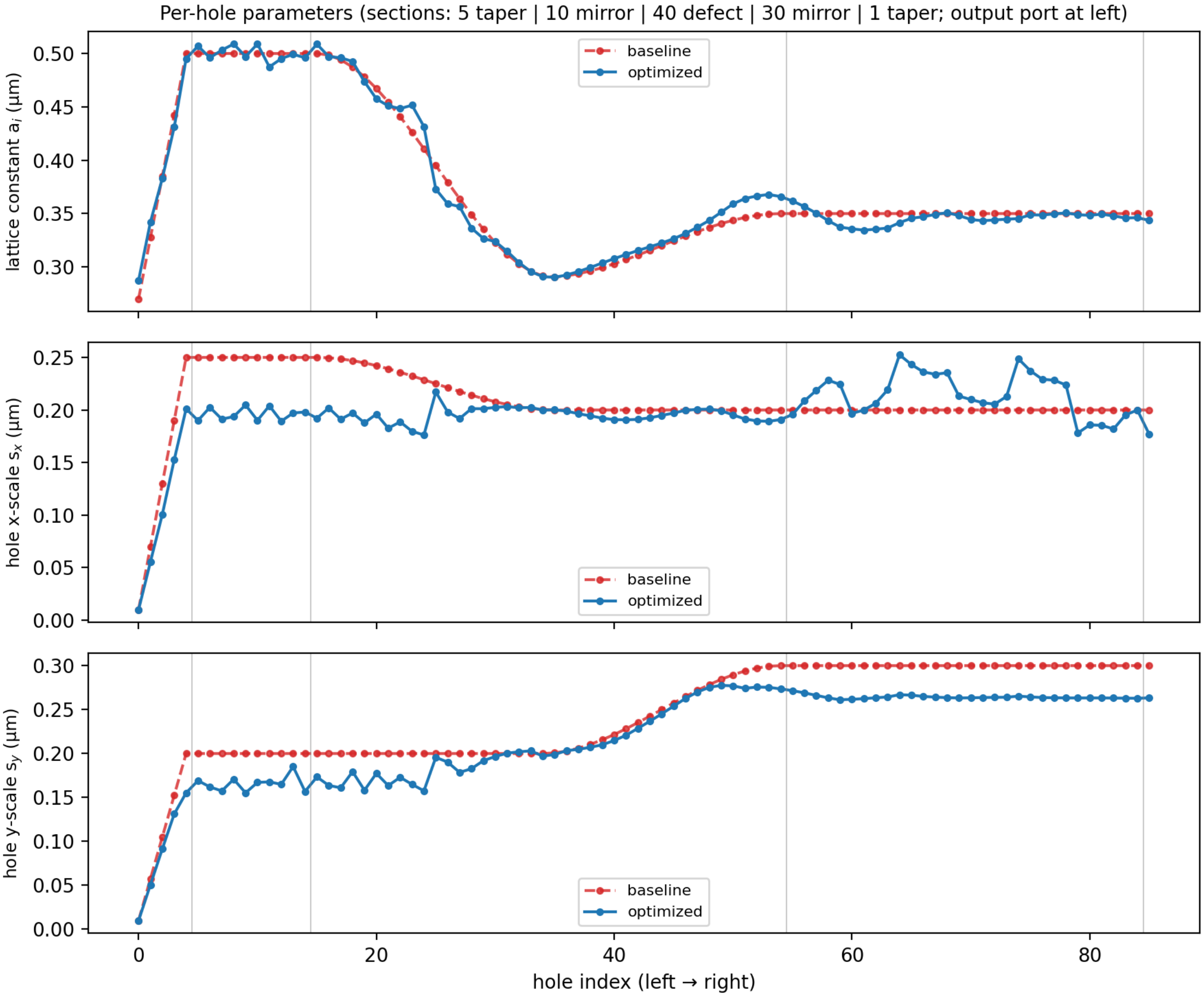

Fig. 3 — Per-hole parameters, baseline (red dashed) vs. optimized (blue). The optimizer kept the defect-region lattice near 0.29 µm but reshaped the left-mirror region (originally a = 0.5 µm — not a working mirror) into a graded structure, and softened the size discontinuities at section boundaries into smooth tapers (gentle confinement).

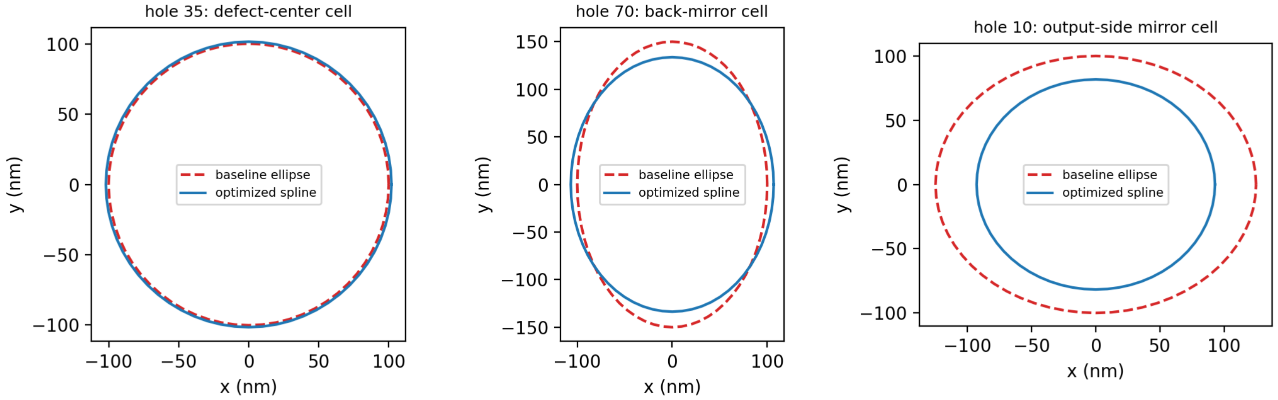

Fig. 4 — Closed-spline hole boundaries vs. baseline ellipses for three representative cells. Shape deformations are modest (control radii within ±3% of circular): at these learning rates the spacing/size schedule, not the hole shape, was the dominant lever.

3. Band structure of the constituent unit cells

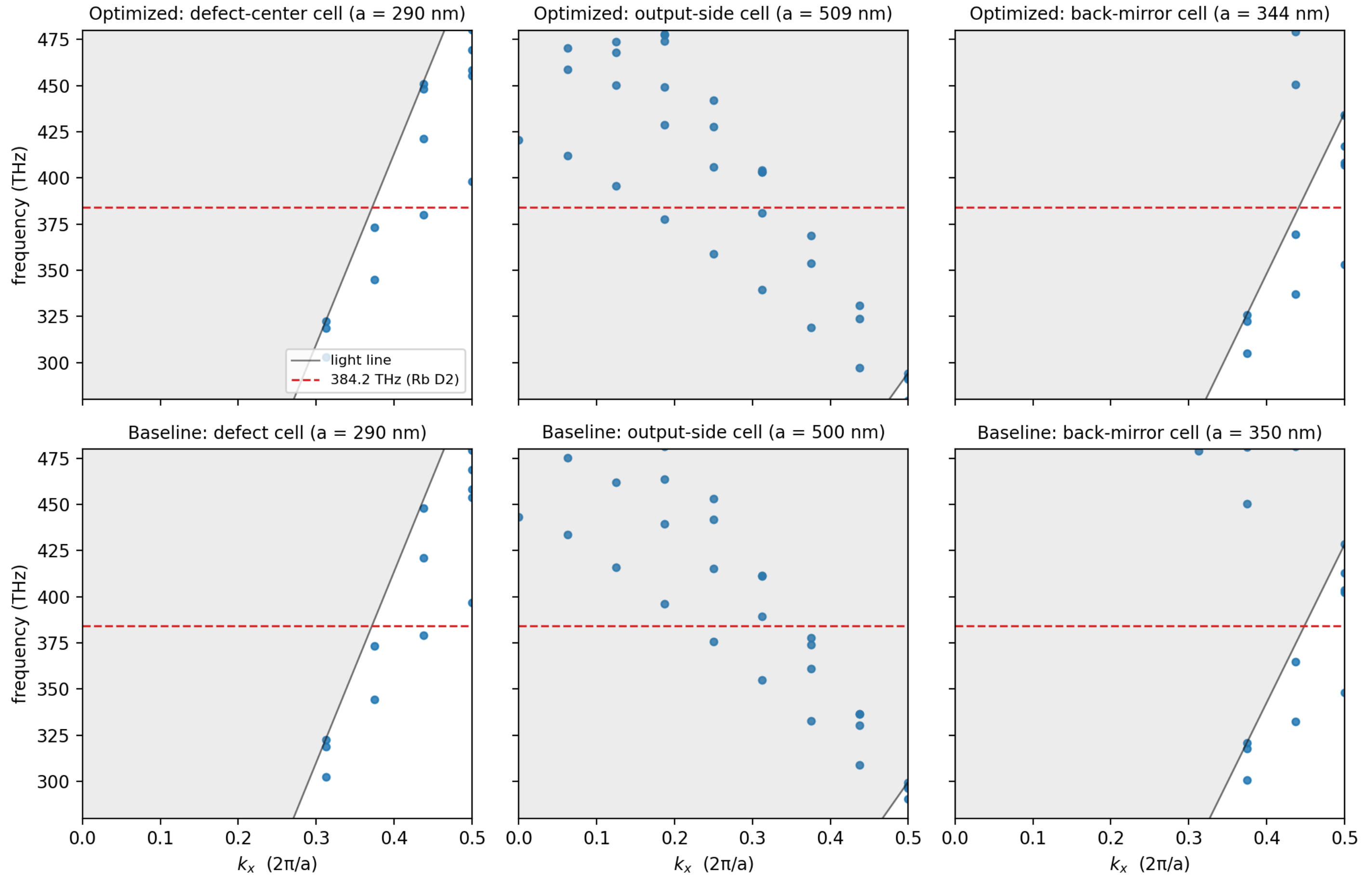

Fig. 5 — TE-like band structures of six constituent unit cells (Tidy3D Bloch unit-cell simulations of the suspended beam, dipole excitation + resonance analysis; MPB is unavailable in this environment). Gray: light cone; red dashed: 384.23 THz Rb D2 target.

The cavity works exactly as the quasipotential picture (Knall et al., PRL 129, 053603 (2022)) prescribes. In the defect cell the dielectric band edge at the zone boundary sits at 398 THz — the 383.9 THz resonance lies below it, in the defect's propagating dielectric band. In the back-mirror cell the gap spans 353–407 THz, so the resonance sits near mid-gap where mirror strength is maximal — light is evanescent there and the back side reflects well (verified Q_{+x} = 2.7×10^5). The baseline's equivalent cells are nearly identical (defect edge 397 THz, back-mirror gap 348–402 THz): the unit-cell photonics was never the baseline's problem.

The output-side cells are the story. With a ≈ 0.5 µm, all guided bands at the zone boundary lie below ≈ 295 THz and the light line at X is c/2a ≈ 300 THz: at 384 THz this section has no band gap and no guided modes — it is not a mirror but a radiative scatterer, in BOTH designs. This is why the baseline lost 91% of its photons to scattering (β = 0.087, Q_y ≈ 940 per face). The optimizer, rather than rebuilding those cells, constructed a smooth graded barrier from the defect-adjacent cells (the lattice descent 0.50 → 0.29 µm visible in Fig. 3, indices 15–27) — the gentle-confinement mechanism of Quan & Lončar (2011) / Tanaka et al. (2008): a gradually rising local band edge confines the mode with minimal Fourier content inside the light cone. That raised β to 0.287 and Q to 1778; the residual a = 0.5 µm segment still radiates and remains the dominant loss channel.

4. Cavity mode fields

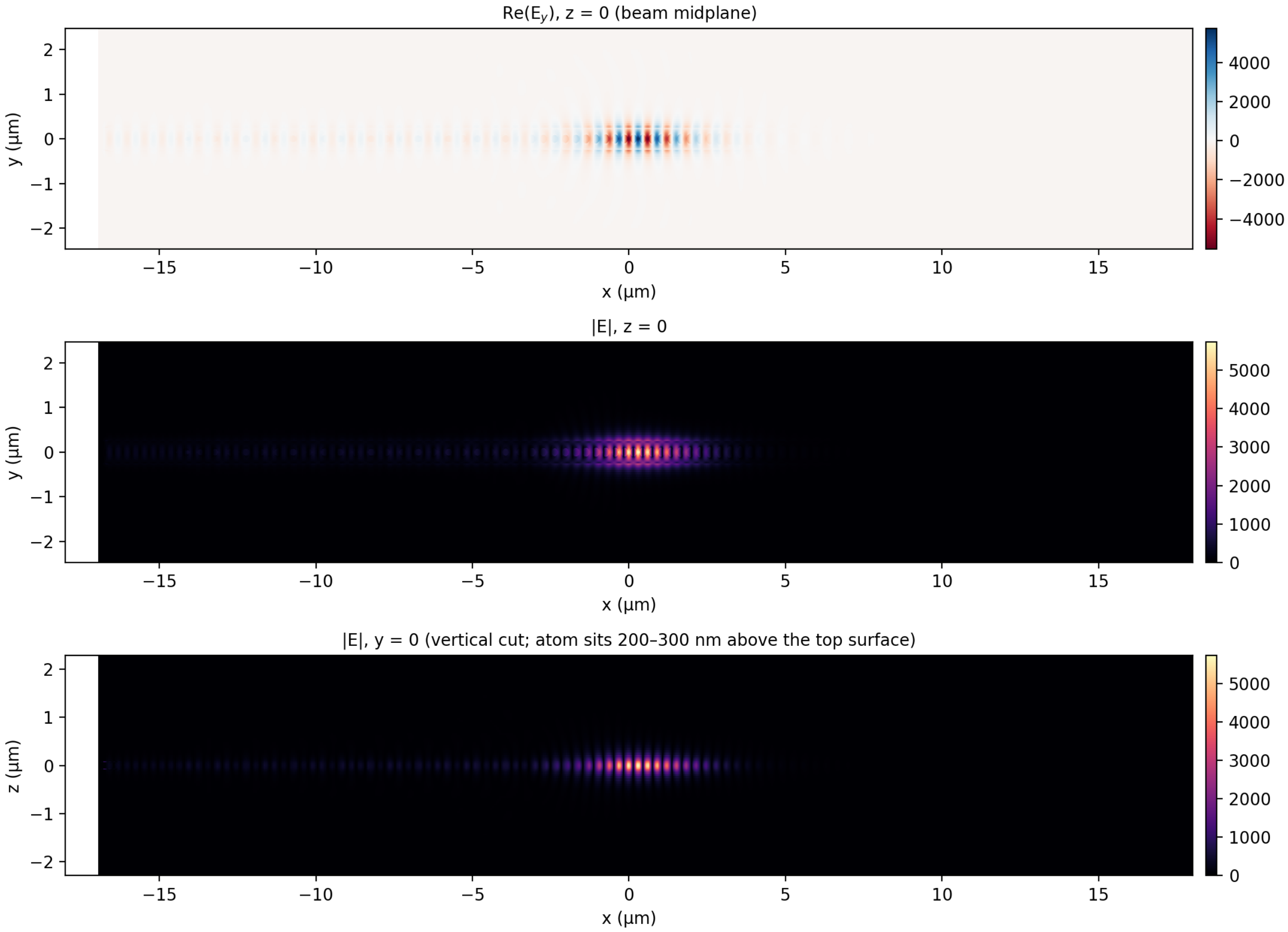

Fig. 6 — Verified cavity mode of the optimized design (final full-quality time-domain simulation, last time step). Top: Re(Ey) at z = 0 showing the Bloch-wave standing pattern under a localized envelope; middle: |E| at z = 0; bottom: |E| in the vertical cut — the evanescent tail above the top surface is what couples to the tweezer-trapped atom at 200–300 nm.

At the atom positions the verified coupling is $g/2\pi$ = 5.38 / 3.84 / 2.78 GHz at 200/250/300 nm — the ~1.9× decay over 100 nm quantifies the design's sensitivity to the tweezer trap distance (set by beam thickness via the retro-reflection phase; Thompson et al. 2013).

5. Optimization trajectory

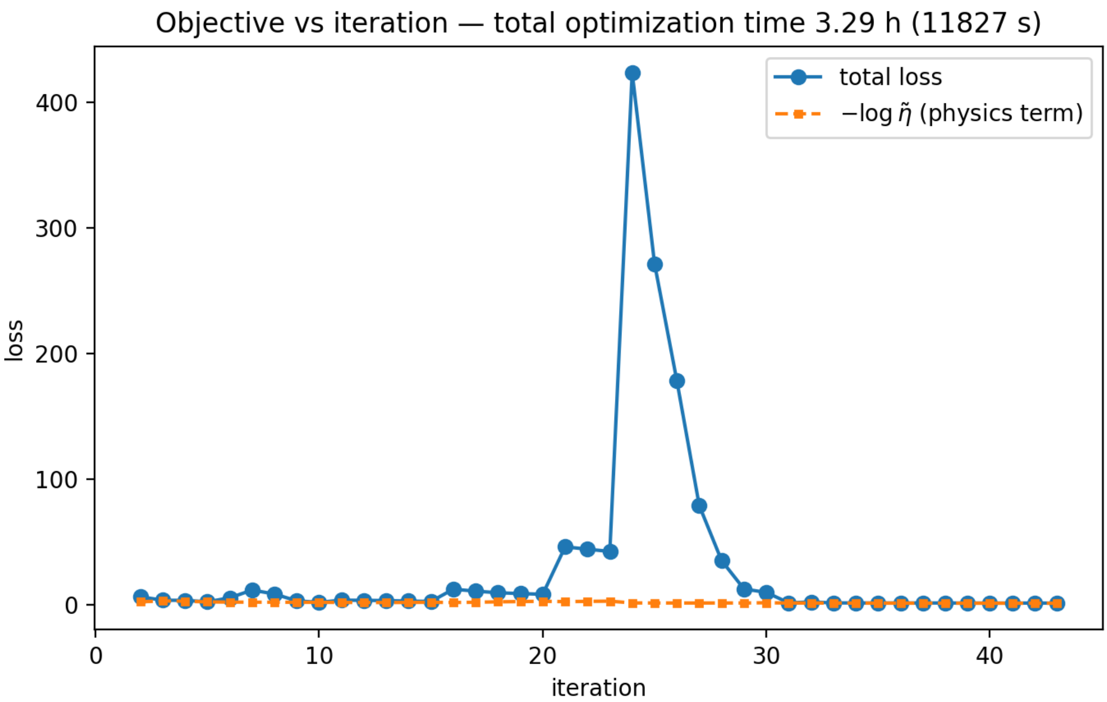

Fig. 7 — Loss vs. iteration. The spike at iteration 24 is the resonance-targeting penalty unsaturating when the monitor window was widened (see text).

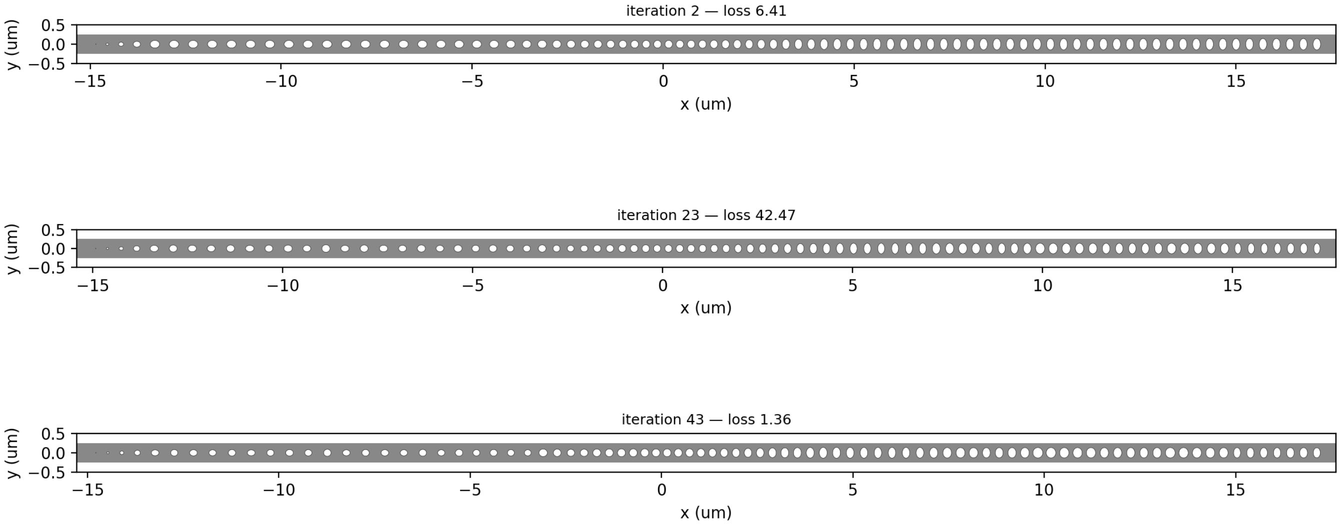

Fig. 8 — Structure evolution (full animation: invDesResults/full/structure_evolution.gif).

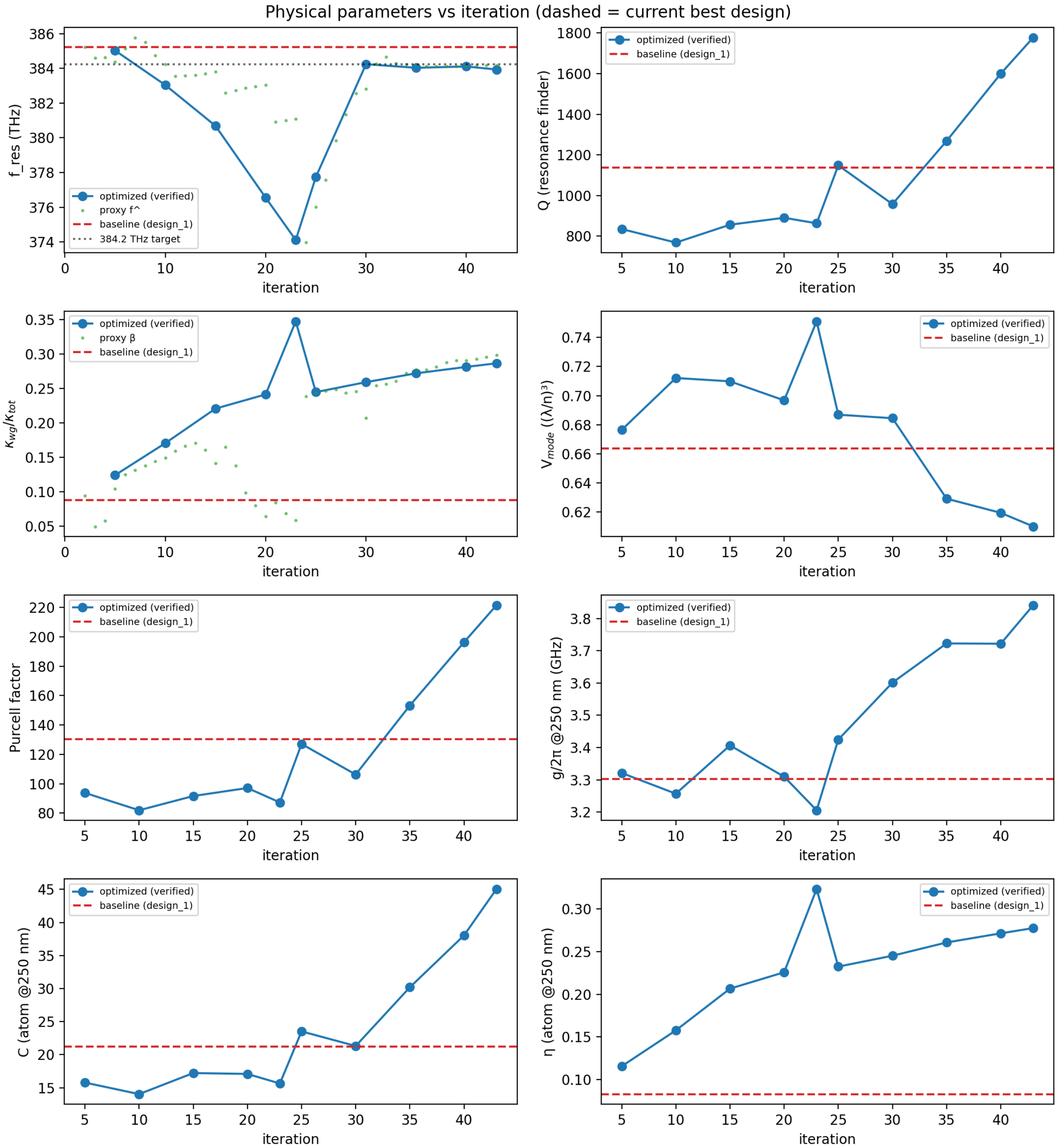

Fig. 9 — Verified physical parameters vs. iteration; dashed red: baseline.

Two-stage schedule: Stage A (iterations 2–23) optimized spacings and hole scales with circular spline control points; Stage B (24–43) unlocked the spline shapes. During Stage A the η term walked the resonance 11 THz below target — with a narrow 2 THz monitor window the softmax frequency estimator saturates at the window edge and the resonance penalty loses its gradient. Widening the window to 6 THz and anchoring it to cover both the verified mode and the target restored the penalty: iterations 24–31 brought the resonance back to 384.2 THz while η continued to rise — the optimizer found designs that are simultaneously on-target and better-coupled.

6. Physical impact in the context of the literature

Atom–photon entanglement fidelity. For time-bin atom–photon entanglement through a nanophotonic cavity the infidelity scales as $\ln C/C$ (Menon et al., NJP 22, 073033 (2020)). The improvement $C = 21 \rightarrow 45$ reduces this error bound from ≈14% to ≈8.5% ($F \approx 0.9$ requires $C \gtrsim 10$–15; demonstrated single-atom records are $C \approx$ 27–71, Suleymanzade & Dimitrova 2025 review). The optimized design's cooperativity at a realistic 250 nm trap position is therefore in the regime of the best demonstrated atom–nanophotonic interfaces, whereas the baseline sat at the threshold.

Remote-entanglement rate. The collection efficiency η enters two-node entanglement rates quadratically (both photons must be collected): $\eta = 0.083 \rightarrow 0.278$ is a 11× speed-up of heralded Bell-pair rates — the difference between sub-kHz and multi-kHz operation at typical attempt rates, directly addressing the >10 kHz remote-entanglement target highlighted in the Suleymanzade & Dimitrova review.

Bad-cavity regime. The hierarchy $\kappa/2\pi = 216$ GHz $\gg g/2\pi = 3.8$ GHz $\gg \gamma/2\pi = 6.1$ MHz places the device deep in the fast/bad-cavity Purcell regime that the Reiserer & Rempe (RMP 2015) and Knall et al. (PRL 2022) analyses favor for broadband, fast photon extraction: protocol bandwidth is set by κ, and the Duan–Kimble reflection gate mechanism is robust to the exact g in this regime.

Where the design still falls short. (i) Outcoupling: $\beta = \kappa_{wg}/\kappa_{tot} = 0.287$ — better than the 10–40% measured in the closest predecessor platform (Menon thesis 2025) but below its >60% goal, and far below the Knall-optimal overcoupling ($\kappa_{wg,opt}/2\pi = \sqrt{\kappa_s(4g^2+\kappa_s\gamma)/\gamma}/2\pi \approx 1.2$ THz $\gg \kappa_s/2\pi = 154$ GHz, i.e. the source-efficiency optimum wants $\beta \rightarrow \approx 0.9$): scattering still carries ~71% of the photons. (ii) Loaded Q = 1778 is orders below the $Q \sim 10^4$–10^5 that gentle-confinement designs reach (Quan & Lončar 2011; Bazin et al. 2014); raising intrinsic Q would raise C and β simultaneously. Both point the same way: this seed design family (the reproducible but weak design_1) limits the achievable basin — the machinery, which is seed-agnostic, should next be run from a band-structure-designed seed (β was still rising when the iteration budget ended; resume from invDesResults/full/checkpoints/latest.pkl).

Caveats on absolute numbers. g and V are quoted via the repo's legacy conventions (time-averaged energy-density ratios from the verification sims); the legacy V_mode includes a symmetry double-count kept for continuity, and g is evaluated on the beam axis directly above the defect. Absolute values should be treated as ~factor-2 estimates; the baseline-to-optimized ratios are computed identically on both designs and are robust.

7. Reproducibility

Record: invDesResults/full/record.json — 42 entries with keys 'params' (full reconstruction state), 'loss', 'time', 'perf'; geometry round-trip verified. Checkpoints: invDesResults/full/checkpoints/ (per-iteration resume state). Commands (tidyEnv, repo root): smoke test 'python -m inverse_design.run_optimization --mode smoke'; full run / extension 'python -m inverse_design.run_optimization --mode full --run-name full --resume'; band diagrams 'python -m inverse_design.bands'; this report 'python -m inverse_design.make_report'. Unit tests: 'pytest tests/' (19 pass). GDS of the optimized design: invDesResults/full/final_design.gds.